# Adaptive Suppression and Enhancement in Auditory Search


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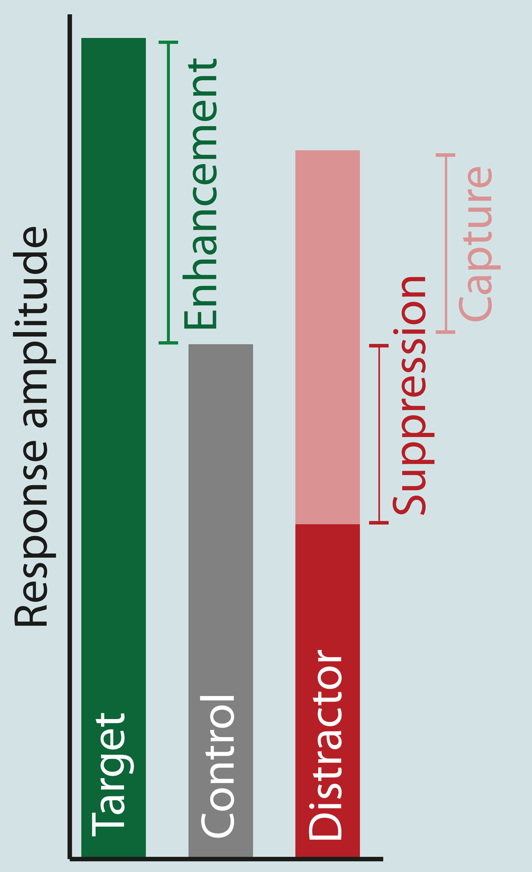
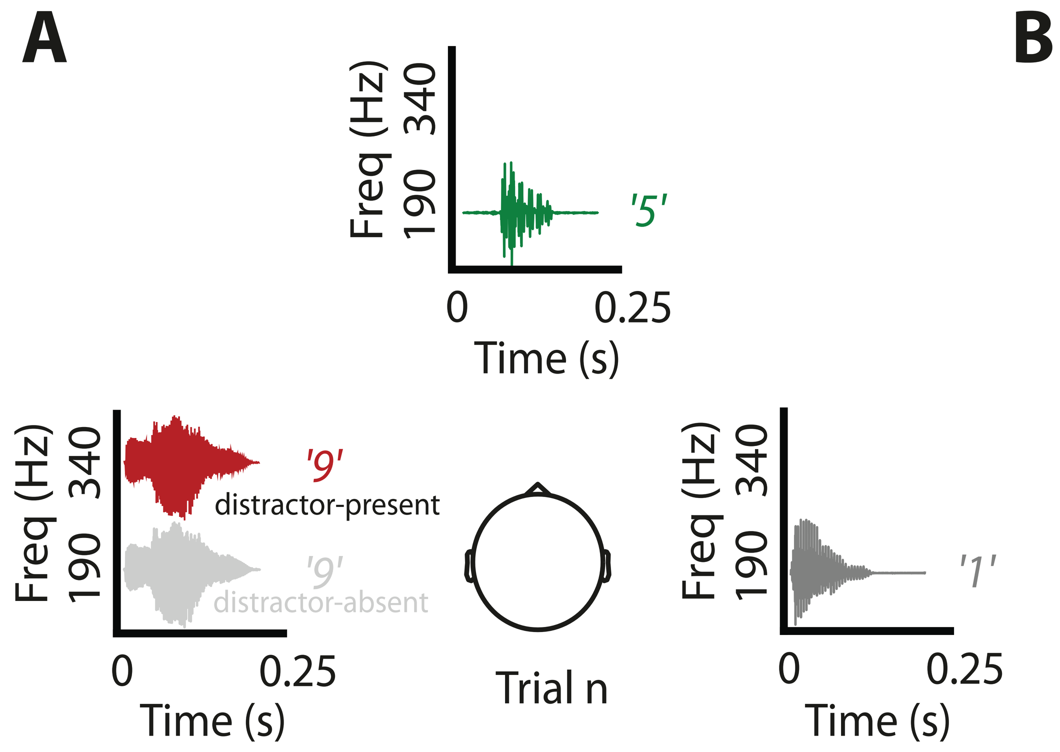
## Background

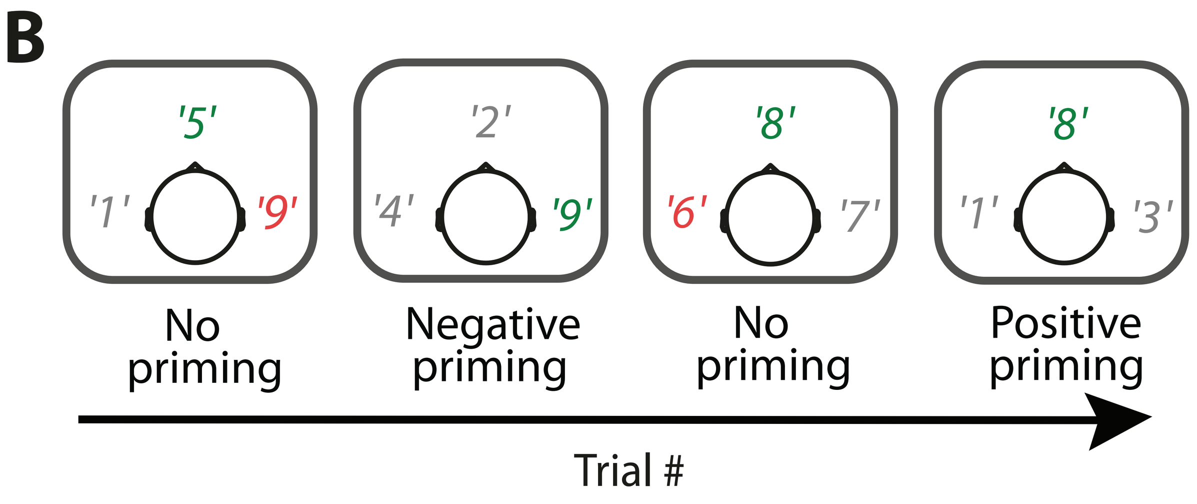
Human environments consist of relevant target and irrelevant distractors. In auditory attention research, the understanding of capture and suppression is premature, partly because target and distractor effects are often confounded[1].

**Research goal**: We introduce a baseline to directly compare neural and behavioral responses between control versus target and distractor sounds, inferring mechanisms of target **enhancement**, distractor **suppression** and **capture**[2].




## Target towardness and Design



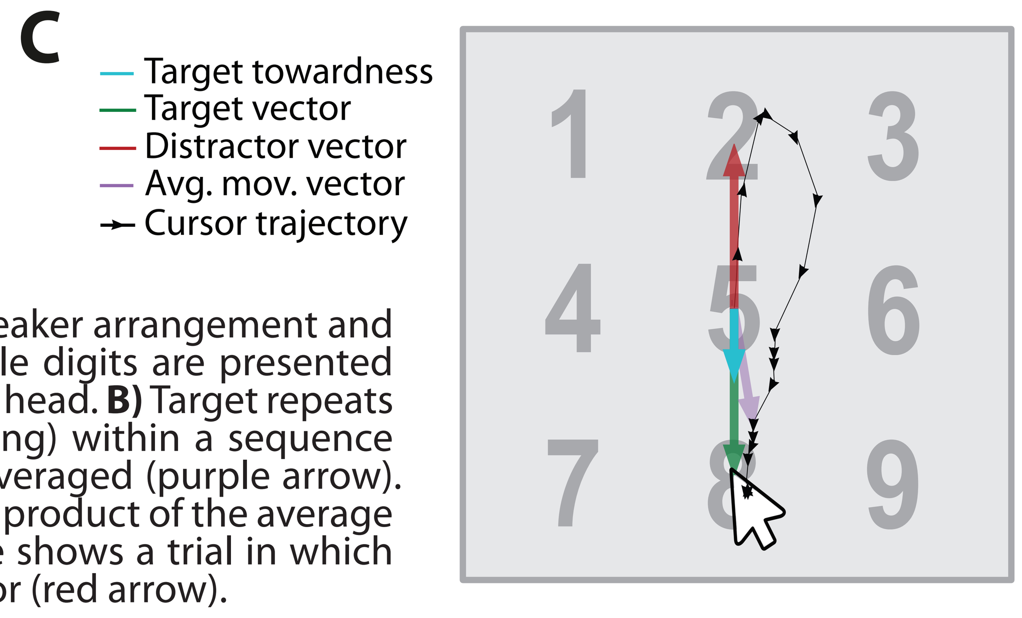




**Figure 1:** Experiment design and cursor trajectory analysis. **A)** Speaker arrangement and categorical sound characteristics. Three simultaneous one-syllable digits are presented from spatial positions -90°, 0°, and +90° relative to the participants' head. **B)** Target repeats (positive priming) and distractor-target switches (negative priming) within a sequence of four consecutive trials[3]. **C)** Trial-wise cursor trajectories were averaged (purple arrow). Target cursor towardness (blue arrow) was calculated as the scalar product of the average movement vector and the target vector (green arrow). The figure shows a trial in which the initial movement direction aimed towards the salient distractor (red arrow).

## Results

### Target Enhancement

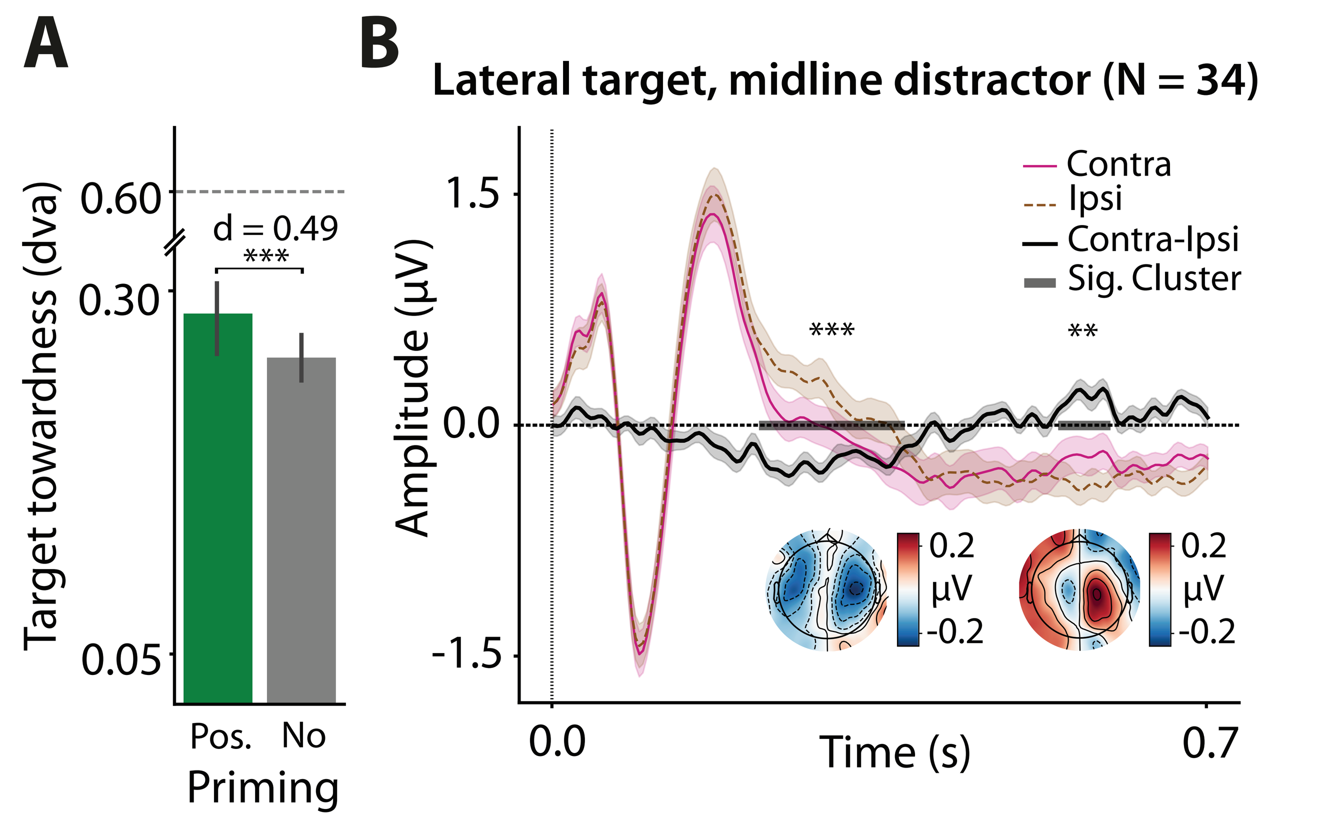



**Figure 2:** Signatures of target enhancement. **A)** Behaviorally, target enhancement was expressed as heightened target towardness in target-repeat trials ($p < 0.001$). Dashed grey line represents optimal performance. **B)** In the EEG, target enhancement was observed by N2ac emergence in lateralized target trials at electrode pair C3/4 ($p = 0.001$). A later component was observed ($p = 0.006$), possibly reflecting spatial reorienting[4,5]. Inset topographies show difference wave distribution during significant time intervals.

### Distractor Suppression

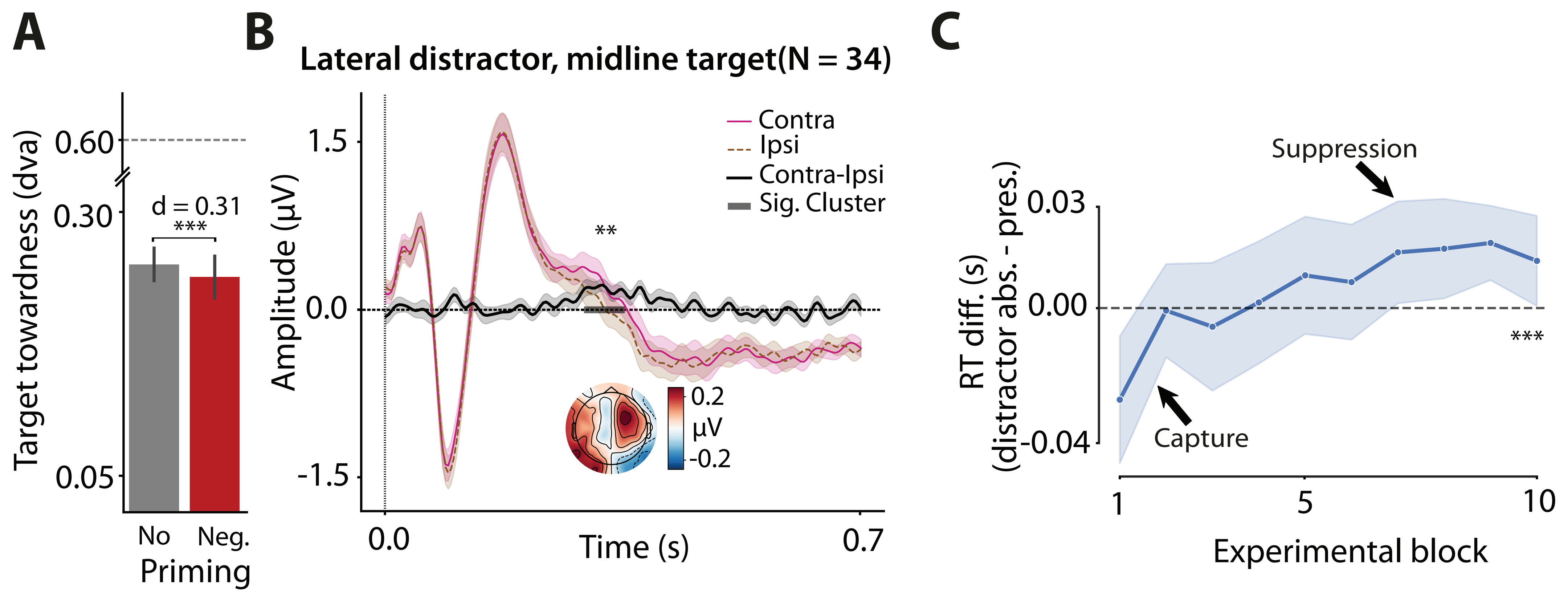



**Figure 3:** Signatures of distractor suppression. **A)** Behaviorally, distractor suppression was expressed as decreased target towardness in distractor-target-switch trials ($p < 0.001$). Dashed grey line represents optimal performance. **B)** In the EEG, distractor suppression was observed by Pd emergence in lateralized distractor trials at electrode pair C3/4 ($p = 0.004$). Inset topographies show difference wave distribution during the significant time interval. **C)** Over the course of the experiment, performance in trials with distractors relatively increased, indicating a gradual transition from attentional capture to more successful distractor suppression (Page's trend test: $L = 10997$, $p < 0.001$).

## Neural Activity Modulates Target Towardness

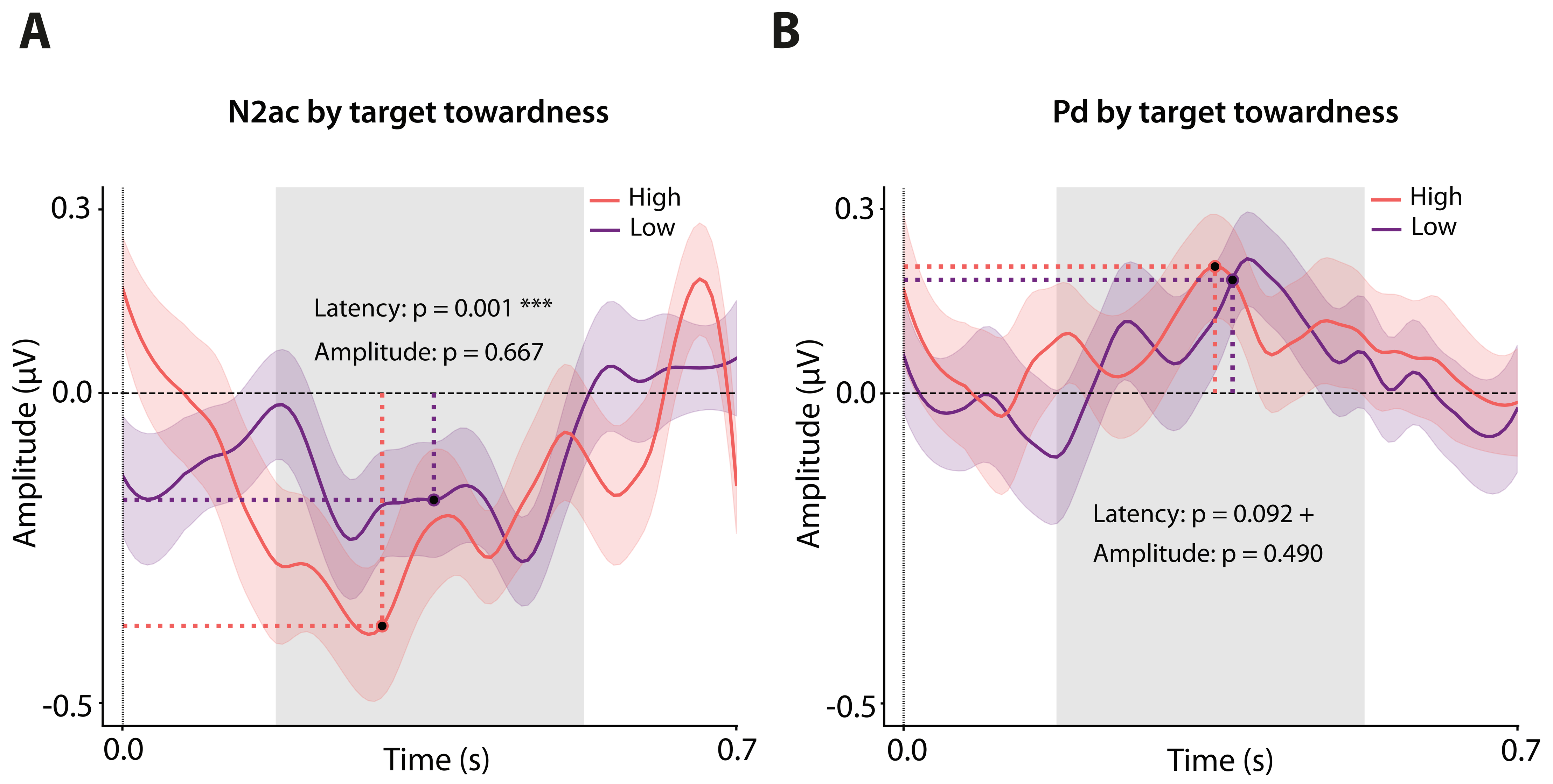



**Figure 4:** Target towardness is modulated by EEG components. Note that both panels visualize identical analysis on lateralized target (N2ac) and distractor (Pd) trials, respectively. Single-trial component latencies and amplitudes were retrieved between 0.2 - 0.4 s post-stimulus onset as the absolute 50% cumulated voltage sum. P values were estimated by a linear mixed-effect model and equated target_towardness ~ (1|Subject ID) + trial number + amplitude + latency. The median-split target towardness serves solely illustration purpose. **A)** Small N2ac latencies evoke greater target towardness compared to large N2ac latencies. The same trend can be observed for lateralized distractor trials in **B)**.

## Conclusion

1. Our design is a novel approach that draws the extensively studied visual search paradigm into the auditory modality.
2. We successfully delineate mechanisms of target enhancement from distractor suppression.
3. The salient distractor initially captures attention, but is gradually suppressed[6].
4. Additionally, we establish a novel metric into auditory search, namely target towardness. This measure effectively captures the above mentioned dynamics of selective attention, negating limitations of conventional metrics.
5. Functionally, rapid neural activity yields optimal search performance.

## References


[1] Wöstmann et al. (2022), Progress in Neurobiology.

[2] Orf et al. (2023), iScience.

[3] Tipper (1985), The Quartlerly Journal of Experimental Psychology.

[4] Gamble & Luck (2011), Psychophysiology.

[5] Gamble & Woldorff (2015), Cerebral Cortex.

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